

Learning Massively Multitask World Models for Continuous Control

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The big picture

Online RL can scale to hundreds of continuous-control tasks when combined with a massively multitask benchmark, a language-conditioned world model, and demonstration-based pretraining.

Motivation

- In previous work, continuous-control RL is still mostly **single-task** or **offline**
- Imitation-heavy scaling is limited by **data quantity** and **demonstration quality**

- Can one agent be trained online across hundreds of tasks?



Motivation

- **Common Issues and Solutions:**
 - **Job Failures:**
 - **Issue:** Job fails or doesn't start.
 - **Solution:** Check resource requests in the script and ensure you have sufficient quota. Review job logs for specific error messages.
 - **Connection Problems:**
 - **Issue:** Unable to log in or connect to Morning Star.
 - **Solution:** Verify your network connection and login credentials. Contact support if the issue persists.
 - **Package Errors:**
 - **Issue:** Errors related to missing or incompatible packages.
 - **Solution:** Ensure you are using the correct virtual environment and that all required packages are installed.

Getting help (II)

- **Where to Find Documentation:**
 - Refer to [SLURM documentation](#) for job scheduling and submission details.
 - Explore the [Digital Research Alliance of Canada's \(the Alliance\) wiki](#) for additional tips and best practices on supercomputing.
- **How to Contact Support:**
 - Reach out to the IT support team through the uOttawa Self-Service Centre (<https://topdesk.uottawa.ca/tas/public/ssp/>) for assistance and troubleshooting.

Best Practices for Using Morning Star (I)

- **Efficient Resource Usage:**
 - **Optimize Jobs:**
 - Tailor your job scripts to request only the resources you need (e.g., time, GPUs).
 - **Monitor Usage:**
 - Regularly check your resource utilization and adjust job requests accordingly
 - Please cancel the interactive window job (with **scancel**) when you are done, do not just close the browser window.
- **Keeping Your Environment Clean and Organized:**
 - **Manage Files:**
 - Regularly clean up unnecessary files and manage your storage quota effectively.
 - **Organize Projects:**
 - Structure your directories logically and use descriptive filenames for easy navigation.

Best Practices for Using Morning Star (II)

- **Important:**
 - **Test Your Script:**
 - Before submitting with 'sbatch', ensure that your script runs correctly in the terminal with a small sample of data. This helps catch any errors early and prevents wasting resources on jobs that might fail.

Wrapping Up

- **Summary of Key Points:**
 - Accessing Morning Star and its features.
 - Submitting and managing compute jobs.
 - Utilizing virtual environments and managing files.
 - Troubleshooting common issues and seeking support.
- **Encouragement to Explore and Experiment:**
 - Take advantage of the powerful resources available on Morning Star.
 - Experiment with different tools and techniques to enhance your research and learning.
- **Invitation for Questions and Further Learning:**
 - Feel free to ask questions or seek clarification on any topics covered.
 - Explore additional resources and documentation for deeper insights and advanced usage.